

Let me run all 5 practicals and then give you a full examiner-ready explanation. All 5 practicals ran perfectly. Now here is your complete examiner-ready guide:

DSBDA Practical Exam — Complete Explanation Guide

PRACTICAL 1 — Titanic Dataset: Data Preprocessing & Normalization

What This Practical Does

This practical loads the Titanic passenger dataset and performs end-to-end data preprocessing — handling missing values, removing irrelevant columns, normalizing numeric data, and encoding categorical variables.

How to Explain the Output Step by Step

Step 1 — Loading & First 5 Rows (`df.head()`)

"Sir/Ma'am, first I loaded the Titanic dataset using `pd.read_csv()`. The `head()` function shows the first 5 rows. The dataset has 891 rows and 12 columns including PassengerId, Survived, Pclass, Sex, Age, Fare, Cabin, Embarked etc."

Step 2 — Missing Values (`df.isnull().sum()`)

"The output showed: Age had 185 missing values, Cabin had 636 missing values (which is about 71% — very high), and Embarked had 42 missing values. This tells us that Cabin is not useful as a feature."

Step 3 — `describe()` output

"The `describe()` function gives statistical summary — count, mean, standard deviation, min, max, and quartile values for all numeric columns. For example, mean Age was around 39 years and mean Fare was 252."

Step 4 — Data Types (`dtypes`)

"The `dtypes` output shows: PassengerId, Survived, Pclass, SibSp, Parch are `int64`. Age and Fare are `float64`. Name, Sex, Ticket, Cabin, Embarked are object (string/categorical)."

Step 5 — Data Cleaning

"I filled missing Age values with the **median** (because median is robust to outliers) and Embarked with the **mode** (most frequent value, which is 'S'). Then I **dropped** Cabin, Name, Ticket, and PassengerId because they don't contribute to analysis."

Step 6 — Min-Max Normalization

"I applied Min-Max Normalization on Age and Fare using `MinMaxScaler`. This scales values to a range of 0 to 1. The formula is: $(x - \min) / (\max - \min)$. This is done so that features with large values like Fare don't dominate features like Age during model training."

Step 7 — Categorical Encoding

"Sex was binary encoded: male=0, female=1. Embarked had 3 categories (C, Q, S) so I used One-Hot Encoding using `pd.get_dummies()` with `drop_first=True` to avoid the dummy variable trap. This created columns Embarked_Q and Embarked_S — if both are False, it means the passenger embarked from C."

Final Shape: (891, 9)

"The final dataset has 891 rows and 9 columns — all numeric now, ready for machine learning."

? Counter Questions & Answers

Q: Why did you use median for Age instead of mean?

A: Median is used because Age may have outliers (e.g., very old passengers). Median is not affected by extreme values, making it more robust than mean.

Q: Why drop Cabin?

A: Cabin had 636 out of 891 values missing — that's about 71% missing. Imputing such a large proportion would introduce too much artificial data and bias the results.

Q: What is Min-Max Normalization? What is its formula?

A: It scales data between 0 and 1. Formula: $X_{\text{scaled}} = (X - X_{\min}) / (X_{\max} - X_{\min})$. It is used so all features are on the same scale.

Q: What is the dummy variable trap and how did you avoid it?

A: When you one-hot encode n categories, you only need n-1 columns. The last category is implied. We use `drop_first=True` to drop one column and avoid multicollinearity.

Q: Why is Sex encoded as 0/1 and not one-hot encoded?

A: Because Sex is binary (only male/female), simple label encoding (map) is sufficient. One-hot encoding is needed when there are 3 or more categories.

✅ PRACTICAL 2 — Academic Performance: Missing Values, Outliers & Transformation

What This Practical Does

Creates a student dataset with intentionally injected missing values, inconsistencies, and outliers, then cleans them step by step and applies log transformation.

How to Explain the Output

Step 1 — Raw Dataset

"Sir, I created a dataset of 50 students with columns like Age, Attendance, MathScore, SciScore, StudyHrs, and Gender. I intentionally added errors to simulate real-world messy data."

Step 2 — Missing Values

"The output showed: Age had 3 missing values, Gender had 2, Attendance had 2, MathScore had 3. These were detected using `isnull().sum()`."

Step 3 — Inconsistencies

"MathScore of 150 was found (impossible since max is 100), and Attendance of -10 was found (negative attendance is impossible). I replaced both with NaN and then handled them."

Step 4 — Filling Missing Values

"Age was filled using **median**, MathScore using **mean** (rounded to 1 decimal), Attendance using median, and Gender using **mode** since it is categorical."

Step 5 — Outlier Detection using IQR

"I used the IQR (Interquartile Range) method. $IQR = Q3 - Q1$. Any value below $Q1 - 1.5 \times IQR$ or above $Q3 + 1.5 \times IQR$ is an outlier. Found: StudyHrs = 22 (outlier) and SciScore = 4 (outlier)."

Step 6 — Handling Outliers — Winsorization (Capping)

"Instead of deleting outliers, I used **capping (Winsorization)** — values above upper bound are set to upper bound, and values below lower bound are set to lower bound. StudyHrs was capped at 11.22."

Step 7 — Log Transformation

"StudyHrs had a skewness of 0.2365 before transformation. After applying `log1p()` (which is $\log(1+x)$), skewness changed to -0.711. This makes the distribution closer to normal, which is required by many ML algorithms."

? Counter Questions & Answers

Q: What is IQR and how is it calculated?

A: $IQR = Q3 - Q1$. Lower bound = $Q1 - 1.5 \times IQR$. Upper bound = $Q3 + 1.5 \times IQR$. Values outside this range are outliers.

Q: Why use Winsorization instead of deleting outliers?

A: Deleting rows means losing data. Winsorization keeps all rows but limits extreme values to the boundary — preserving dataset size while removing the impact of outliers.

Q: Why use `log1p` instead of `log`?

A: `log(0)` is undefined (negative infinity). `log1p(x) = log(1+x)` safely handles zero values, making it more robust.

Q: Why fill numeric columns with mean/median but categorical with mode?

A: Mean/median are meaningful for numbers. For categories like Gender, you can't take a mean — so you fill with the most frequently occurring value (mode).

Q: What does skewness value tell us?

A: Skewness = 0 means perfectly symmetric. Positive skewness means right tail is longer. Negative skewness means left tail is longer. Values between -0.5 and 0.5 are considered approximately normal.

✓ PRACTICAL 3 — Statistical Analysis: Custom Dataset + Iris Dataset

What This Practical Does

Part 1 creates a custom student score dataset and computes grouped statistics. Part 2 uses the famous Iris dataset and computes detailed statistics per species.

How to Explain the Output

Part 1 — Custom Dataset

"I created a dataset of 25 students with 4 columns: Student name, Grade_Group (A/B/C), Score (out of 100), and StudyHrs per day. I saved it as a CSV file and loaded it back."

Grouped Statistics — Score by Grade Group:

"Grade A students had a mean score of 90, median 90, min 85, max 95, std 3.12. Grade B had mean 75.5. Grade C had mean 64.38. This clearly shows higher grade groups have higher scores."

Grouped Statistics — StudyHrs by Grade Group:

"Grade A students studied 5.67 hours on average, Grade B 3.5 hours, Grade C only 2.12 hours. This shows a direct correlation between study hours and performance."

Part 2 — Iris Dataset

"The Iris dataset from sklearn has 150 rows and 5 columns: sepal length, sepal width, petal length, petal width, and species. There are 3 species: setosa, versicolor, virginica — 50 samples each."

Key statistics to mention:

- Setosa: smallest petals (mean petal length 1.46 cm, width 0.25 cm), most compact species
- Versicolor: medium (mean petal length 4.26 cm)
- Virginica: largest (mean petal length 5.55 cm, width 2.03 cm)

"I computed describe(), percentiles (10th, 25th, 50th, 75th, 90th), mean, and standard deviation for each species separately."

? Counter Questions & Answers

Q: What does `groupby()` do?

A: It groups rows by a column's unique values and allows you to apply aggregation functions (mean, sum, count etc.) on each group separately.

Q: What is the difference between mean and median?

A: Mean is the arithmetic average — sensitive to outliers. Median is the middle value — robust to outliers. For skewed data, median is preferred.

Q: What is standard deviation?

A: It measures the spread of values around the mean. A low std means values are close to the mean (consistent). A high std means values are more spread out (variable).

Q: What are percentiles?

A: The p-th percentile is the value below which p% of the data falls. 25th percentile = Q1, 50th = median (Q2), 75th = Q3.

Q: Why is the Iris dataset commonly used?

A: It is a clean, balanced, well-understood dataset with 3 classes and 4 numeric features — ideal for learning classification, statistics, and visualization.

✅ PRACTICAL 4 — Linear Regression: House Price Prediction

What This Practical Does

Builds a Linear Regression model to predict house prices using features like rooms (RM), % lower status population (LSTAT), pupil-teacher ratio (PTRATIO), tax rate (TAX), and crime rate (CRIM).

How to Explain the Output

Step 1 — Dataset

"I created a dataset of 20 houses with 5 features and 1 target variable (HousePrice). The dataset had missing values in RM, LSTAT, and PTRATIO which were filled using column mean."

Step 2 — Feature Scaling

"I used StandardScaler which standardizes features to mean=0 and std=1. Formula: $Z = (X - \text{mean}) / \text{std}$. This ensures all features contribute equally to the regression."

Step 3 — Train-Test Split

"80% of data (16 samples) was used for training and 20% (4 samples) for testing using `train_test_split` with `random_state=42` for reproducibility."

Step 4 — Model Coefficients

"The model learned these coefficients: RM=+0.484 (more rooms → higher price), LSTAT=+0.368, PTRATIO=-1.220 (higher ratio → lower price), TAX=-13.810 (higher tax → lower price), CRIM=+7.954. Intercept=25.18."

Step 5 — Actual vs Predicted

"For the 4 test houses: Actual prices were 25, 39, 14, 32. Predicted were 20.84, 37.71, 13.94, 30.64 — very close, showing the model learned well."

Step 6 — Evaluation Metrics

"MAE = 1.718 (average error of ₹1.72 thousand), MSE = 5.212, RMSE = 2.283, and $R^2 = 0.939$ — meaning the model explains **93.9% of the variance** in house prices. This is an excellent result."

? Counter Questions & Answers

Q: What is Linear Regression?

A: It models the relationship between one or more independent variables (features) and a continuous dependent variable (target) by fitting a straight line:

$$Y = mX + c.$$

Q: What does R^2 score mean?

A: R^2 (R-squared) tells us what proportion of the variance in the target variable is explained by the model. $R^2=1$ means perfect prediction. $R^2=0$ means the model is no better than just predicting the mean. $R^2=0.939$ means 93.9% of variance is explained.

Q: What is the difference between MAE, MSE, and RMSE?

A: MAE = average of absolute errors (easy to interpret). MSE = average of squared errors (penalizes large errors more). RMSE = square root of MSE (same unit as target, easier to interpret than MSE).

Q: Why use StandardScaler before Linear Regression?

A: Features with large magnitudes (like TAX=420) would dominate over small-magnitude features (like CRIM=0.02). Scaling ensures each feature has equal influence on the model.

Q: What is `random_state` in `train_test_split`?

A: It sets a seed for the random number generator to ensure reproducibility — running the code multiple times gives the same split.

✓ PRACTICAL 5 — Logistic Regression: Purchase Prediction

What This Practical Does

Builds a Logistic Regression model to classify whether a user will purchase a product based on Age and Estimated Salary. Evaluates using confusion matrix and metrics.

How to Explain the Output

Step 1 — Dataset

"I created a dataset of 100 users with UserID, Gender, Age, EstimatedSalary, and Purchased (0 or 1). The purchase logic was: older users with higher salary are more likely to buy."

Step 2 — Class Distribution

"65 users Purchased (class 1), 35 did not (class 0). This is a mildly imbalanced dataset."

Step 3 — Train-Test Split

"75 samples for training, 25 for testing (75-25 split using `test_size=0.25`)."

Step 4 — Confusion Matrix The matrix showed:

	Predicted NO	Predicted YES
Actual NO	TN = 7	FP = 4
Actual YES	FN = 0	TP = 14

"TN=7 means 7 non-buyers correctly predicted as not buying. TP=14 means 14 buyers correctly predicted as buying. FP=4 means 4 non-buyers were wrongly predicted as buyers. FN=0 means no actual buyer was missed — which is excellent."

Step 5 — Evaluation Metrics

"Accuracy = 84% — 21 out of 25 correct predictions. Error Rate = 16%. Precision = 0.7778 — of all predicted buyers, 77.78% actually bought. **Recall = 1.0 — the model caught 100% of actual buyers — zero false negatives.** This is very important in marketing — you don't want to miss any potential buyer."

Classification Report:

"For 'Not Purchased': precision=1.0, recall=0.64. For 'Purchased': precision=0.78, recall=1.0. F1-score for Purchased class is 0.88."

? Counter Questions & Answers

Q: What is the difference between Linear and Logistic Regression?

A: Linear Regression predicts a continuous value (e.g., price). Logistic Regression predicts a binary class (0 or 1) by using a sigmoid function to output probabilities between 0 and 1.

Q: What is a Confusion Matrix?

A: A 2×2 table showing TP (correctly predicted positive), TN (correctly predicted negative), FP (predicted positive but actually negative — Type I error), and FN (predicted negative but actually positive — Type II error).

Q: What is Precision vs Recall?

A: Precision = $TP / (TP+FP)$ — out of all predicted positives, how many are actually positive. Recall = $TP / (TP+FN)$ — out of all actual positives, how many did we capture. In this practical, Recall = 1.0 meaning we captured 100% of buyers.

Q: What is F1-Score?

A: F1-Score = $2 \times (Precision \times Recall) / (Precision + Recall)$. It is the harmonic mean of precision and recall — useful when the dataset is imbalanced and you want a balance between the two.

Q: Why use Logistic Regression instead of Linear Regression for classification?

A: Linear Regression can give values outside 0 and 1 (like -0.3 or 1.7), which doesn't make sense for probability. Logistic Regression uses the sigmoid function to constrain output between 0 and 1, making it suitable for binary classification.

Q: What is the sigmoid function?

A: $\sigma(z) = 1 / (1 + e^{(-z)})$. It takes any real number and maps it to a value between 0 and 1, which is interpreted as the probability of belonging to class 1.

Quick Summary Table

Practical	Topic	Key Output
P1	Titanic Preprocessing	Cleaned data, Min-Max normalization, One-Hot encoding, final shape (891,9)
P2	Outlier & Missing Data	IQR outlier detection, Winsorization capping, Log transform (skewness 0.24→-0.71)
P3	Statistics	Grade-wise grouped stats, Iris species-wise describe(), percentiles, mean, std
P4	Linear Regression	$R^2=0.939$, RMSE=2.283, model explains 93.9% variance
P5	Logistic Regression	Accuracy=84%, Recall=1.0 (zero buyers missed), Confusion Matrix

Pro tip for the exam: Always start with *"Sir/Ma'am, the purpose of this practical is..."*, explain the dataset, then walk through the output section by section. For every metric you show, say what it means in plain English — examiners love that.

Let me run all the Python practicals first, then give you the full guide including the Hadoop Java practical. All outputs captured. Here is your complete examiner-ready guide:

DSBDA Practicals 6–10 + Hadoop — Complete Explanation Guide

✓ PRACTICAL 6 — Naive Bayes Classification on Iris Dataset

What This Practical Does

Loads the Iris dataset, trains a **Gaussian Naive Bayes** classifier to identify 3 flower species, and evaluates using confusion matrix and metrics.

How to Explain the Output Step by Step

Step 1 — Dataset

"Sir/Ma'am, I used the Iris dataset from sklearn — 150 rows, 6 columns (4 features + target + species name). The 3 species are setosa, versicolor, and virginica, with 50 samples each."

Step 2 — Train-Test Split

"120 samples were used for training and 30 for testing — an 80-20 split with `random_state=42`."

Step 3 — Confusion Matrix

	setosa	versicolor	virginica
setosa	10	0	0
versicolor	0	9	0
virginica	0	0	11

"The diagonal values (10, 9, 11) represent correctly classified samples. All off-diagonal values are 0 — meaning **not a single flower was misclassified**. This is a perfect confusion matrix."

Step 4 — TP, FP, FN, TN per Class

"For setosa: TP=10 (correctly predicted as setosa), FP=0 (no other class was wrongly called setosa), FN=0 (no setosa was missed), TN=20 (remaining 20 samples correctly not called setosa). Same perfect scores for all three classes."

Step 5 — Evaluation Metrics

"Accuracy = 1.0 (100%), Error Rate = 0.0, Precision = 1.0, Recall = 1.0. The model achieved **perfect classification** on all 30 test samples."

Classification Report:

"Every class has precision=1.0, recall=1.0, f1-score=1.0. The model perfectly separated all three species."

? Counter Questions & Answers

Q: What is Naive Bayes? Why is it called "Naive"?

A: Naive Bayes is a probabilistic classifier based on Bayes' Theorem. It is called "Naive" because it assumes all features are **independent** of each other, which is a simplifying assumption that rarely holds perfectly in real data.

Q: What is Bayes' Theorem?

A: $P(\text{Class} \mid \text{Features}) = P(\text{Features} \mid \text{Class}) \times P(\text{Class}) / P(\text{Features})$. It calculates the probability of a class given the feature values. The class with the highest probability is the predicted class.

Q: Why GaussianNB specifically?

A: Because the Iris features (sepal/petal measurements) are **continuous numeric values** that roughly follow a normal (Gaussian) distribution. GaussianNB assumes features follow a Gaussian distribution for each class. For discrete data, we would use MultinomialNB.

Q: Why did the model get 100% accuracy on Iris?

A: Iris is a clean, well-separated dataset. Setosa is linearly separable from the other two species. With only 30 test samples and a favorable random split, perfect accuracy is achievable. In practice, real-world datasets rarely give 100%.

Q: How is macro average calculated for Precision/Recall?

A: Macro average = simple average of per-class values. Here, $(1.0 + 1.0 + 1.0) / 3 = 1.0$. It treats all classes equally regardless of how many samples they have.

Q: What is the difference between Precision and Recall?

A: Precision = $TP / (TP + FP)$ — of all predicted positives, how many are actually positive. Recall = $TP / (TP + FN)$ — of all actual positives, how many did the model find. In this practical, both are 1.0 since $FP=0$ and $FN=0$.

✓ PRACTICAL 7 — Text Analysis & NLP Preprocessing + TF-IDF

What This Practical Does

Part 1 preprocesses a text document through 6 NLP steps. Part 2 computes TF-IDF scores for a 4-document corpus.

How to Explain the Output Step by Step

Original Document:

"Sir, I used a 3-sentence paragraph about Natural Language Processing as input."

Step 1 — Sentence Tokenization

"Sentence tokenization splits the document into individual sentences using punctuation like full stops. Our 3-sentence paragraph became 3 separate sentence tokens."

Step 2 — Word Tokenization

"Word tokenization splits sentences into individual words and punctuation marks. For example, 'Machines are learning' becomes ['Machines', 'are', 'learning']. Punctuation like . is also treated as a token."

Step 3 — POS Tagging

"POS (Part-of-Speech) tagging assigns grammatical labels to each word. For example: 'Natural'→JJ (Adjective), 'Language'→NNP (Proper Noun), 'is'→VBZ (Verb), 'learning'→VBG (Verb gerund/present participle)."

Step 4 — Stopword Removal

"Stopwords are common words like 'is', 'a', 'of', 'the', 'are', 'to' that carry no meaningful information. After removal, only content words like 'Natural', 'Language', 'Processing', 'fascinating', 'Machines', 'learning' remain."

Step 5 — Stemming

"Stemming chops words to their root form using rules — not always a real word. Examples: 'Natural'→'natur', 'learning'→'learn', 'techniques'→'techniqu', 'quality'→'qualiti'. PorterStemmer is the algorithm used."

Step 6 — Lemmatization

"Lemmatization also finds the root word but uses a dictionary to return valid words. Examples: 'Machines'→'Machine', 'models'→'model', 'techniques'→'technique'. It's slower but more accurate than stemming."

Part 2 — TF-IDF Manual Calculation

"For the word 'learning' in Doc2 ('Machine learning helps machines learn from data'):"

- $TF = 1/7 = 0.1429$ (appears 1 time in 7 words)
- $IDF = \log(4/2) = 0.6931$ (appears in 2 of 4 documents)

- $TF\text{-}IDF = 0.1429 \times 0.6931 = \mathbf{0.0990}$

"Higher TF-IDF means the word is important to that document but not too common across all documents."

? Counter Questions & Answers

Q: What is the difference between Stemming and Lemmatization?

A: Stemming is a rule-based approach that strips suffixes — fast but can produce non-words ('qualiti', 'natur'). Lemmatization uses a vocabulary dictionary to return actual base forms ('quality', 'natural') — slower but linguistically correct.

Q: What are Stopwords? Give 5 examples.

A: Common words that add no meaning to text analysis: 'is', 'a', 'the', 'are', 'of', 'to', 'in', 'and', 'it'. They are removed to reduce noise and improve model performance.

Q: What is TF-IDF? What is its formula?

A: $TF\text{-}IDF = TF \times IDF$. TF (Term Frequency) = count of word in document / total words in document. IDF (Inverse Document Frequency) = $\log(\text{total documents} / \text{documents containing the word})$. A high TF-IDF means the word is frequent in this document but rare across all documents — so it's distinctive.

Q: Why do we use **log** in IDF?

A: To dampen the effect of IDF. Without log, a word appearing in 1 out of 1000 documents would give $IDF=1000$, which is too large. Log brings it to a manageable scale ($\log(1000) \approx 6.9$).

Q: What is the difference between Word Tokenization and Sentence Tokenization?

A: Sentence tokenization splits text into sentences (using **.?!** as delimiters). Word tokenization splits sentences into individual words and punctuation tokens.

Q: What is POS tagging used for?

A: It helps in understanding the grammatical structure of text — useful in named entity recognition, information extraction, and machine translation. For example, knowing 'learning' is a verb vs a noun changes interpretation.

✓ PRACTICAL 8 — Data Visualization: Titanic Dataset (3 Plots)

What This Practical Does

Creates a 50-passenger Titanic-like dataset and generates 3 visualizations: Histogram, Count Plot, and Scatter Plot.

How to Explain the Output

Dataset:

"Sir, I created a dataset of 50 Titanic passengers with columns PassengerId, Survived (0/1), Pclass (1/2/3), Sex (male/female), Age, and Fare. It had 1 missing Age value and 1 missing Fare, both filled using column mean."

Plot 1 — Histogram of Fare (**histogram_fare.png**)

"The histogram shows the distribution of fare prices across 50 passengers. The X-axis shows fare amount, Y-axis shows count of passengers. The KDE (Kernel Density Estimate) curve overlaid shows the probability density. The distribution appears roughly uniform since the data was randomly generated between 50 and 500."

Plot 2 — Count Plot: Survival by Gender (**survival_countplot.png**)

"This count plot shows the number of survived (1) vs not survived (0) passengers, split by gender using color. It helps visualize whether survival rates differ between male and female passengers. In real Titanic data, females have significantly higher survival rates."

Plot 3 — Scatter Plot: Age vs Fare (**scatter_age_fare.png**)

"Each point represents one passenger. X-axis = Age, Y-axis = Fare, color = Survived (0 or 1). This plot helps identify if older passengers paid higher fares and whether fare/age influenced survival. Since this is random data, no clear pattern is expected."

? Counter Questions & Answers

Q: What is a Histogram and how is it different from a Bar Chart?

A: A histogram shows the frequency distribution of a **continuous** numeric variable by grouping data into bins. A bar chart compares **categorical** data. In a histogram, bars are adjacent (no gaps); in a bar chart, they have gaps.

Q: What is KDE in **sns.histplot(kde=True)**?

A: KDE (Kernel Density Estimate) is a smooth curve that represents the probability density of the data. It gives a continuous approximation of the distribution instead of discrete bins.

Q: What is a Scatter Plot used for?

A: A scatter plot shows the relationship between two continuous variables. Each point represents one observation. It helps identify correlation, clusters, or outliers between two features.

Q: What does `hue` parameter do in seaborn?

A: The `hue` parameter colors the data points or bars based on a categorical variable (e.g., Sex or Survived), making it easy to compare groups visually within the same plot.

Q: What is `np.random.seed(42)` for?

A: It sets a fixed seed for the random number generator so results are reproducible — running the code again gives the same "random" data.

✓ PRACTICAL 9 — Advanced Data Visualization: Titanic (Box Plot, Line Chart, Scatter)

What This Practical Does

Creates a 500-passenger realistic Titanic dataset with gender/class-based survival probabilities, then generates 3 advanced visualizations.

How to Explain the Output

Dataset:

"I created a dataset of 500 passengers. Survival probability was realistically modelled — females had 74% base survival chance, males only 19%. 1st class passengers got +15%, 3rd class got -10%. This gives realistic patterns."

Survival Counts: 329 did not survive (0), 171 survived (1). Gender: 327 male, 173 female.

Plot 1 — Box Plot: Age by Gender and Survival (`p9_boxplot_age_gender.png`)

"A box plot shows the five-number summary: minimum, Q1 (25th percentile), median, Q3 (75th percentile), and maximum. The box covers the IQR (Q1 to Q3), the line inside is the median, and the whiskers extend to 1.5×IQR. Points outside are outliers. Here, we see Age distribution for male/female passengers grouped by survival status. Female survivors tend to be slightly younger. Male non-survivors show a wider age spread."

Plot 2 — Line Chart: Survival Rate by Age Group (`p9_linechart_survival_age.png`)

"Passengers were grouped into age bands: 0-10, 11-20, 21-30... and survival rate (proportion who survived) was calculated for each group. The line chart shows this trend. Younger passengers (0-10) have the highest survival rate —

consistent with 'women and children first' policy. Survival generally decreases with age."

Plot 3 — Scatter Plot: Age vs Fare by Gender & Survival

(p9_scatter_age_fare.png)

"Each point's color represents gender (blue=male, pink=female) and shape represents survival status. Female passengers (pink) paid higher fares on average since they were more likely 1st/2nd class. Higher fare passengers show better survival — visible as a pattern in the upper portion of the plot."

? Counter Questions & Answers

Q: What does a Box Plot show? Explain its components.

A: A box plot (box-and-whisker plot) shows:

- Box = IQR (Q1 to Q3, middle 50% of data)
- Line inside box = Median (Q2)
- Whiskers = extend to $Q1 - 1.5 \times IQR$ and $Q3 + 1.5 \times IQR$
- Points beyond whiskers = Outliers

Q: What is `pd.cut()` used for in this practical?

A: `pd.cut()` divides a continuous variable (Age) into discrete bins (age groups like 0-10, 11-20 etc.), creating a new categorical column called `age_group`. This is called **binning or discretization**.

Q: What is `groupby().mean()` doing in the Line Chart section?

A: It groups the dataset by `age_group`, then computes the mean of `survived` for each group. Since `survived` is 0/1, the mean equals the survival rate (proportion of 1s) for that age group.

Q: Why use `np.random.binomial()` for survival?

A: `np.random.binomial(1, p)` generates 0 or 1 with probability `p` of getting 1. This models a Bernoulli trial — whether each passenger survives (1) or not (0) based on their calculated probability.

✓ PRACTICAL 10 — Data Visualization: Iris Dataset (4 Plots + Outlier Detection)

What This Practical Does

Loads the Iris dataset, renames columns, saves as CSV, generates 4 visualizations (histograms, box plots, line chart, scatter), and detects outliers using IQR.

How to Explain the Output

Dataset:

"The Iris dataset has 150 rows and 5 columns: sepal_length, sepal_width, petal_length, petal_width (all continuous numeric), and species (nominal categorical — setosa, versicolor, virginica)."

Plot 1 — Histograms of All 4 Features (p10_histograms.png)

"4 histograms in a 2x2 grid, one per feature, each with KDE curve. Key observations:

- petal_length and petal_width show **bimodal distribution** — two humps — because setosa has very small petals while versicolor and virginica have larger ones.
- sepal_length shows an approximately normal distribution.
- sepal_width is the most normally distributed feature."

Plot 2 — Box Plots by Species (p10_boxplots.png)

"4 box plots showing how each feature varies across the 3 species. Key observations:

- Setosa has the **smallest** petal_length (mean 1.46 cm) and petal_width (mean 0.25 cm).
- Virginica has the **largest** (mean petal_length 5.55 cm).
- Setosa's sepal_width is actually the highest among the three species."

Plot 3 — Line Chart: Mean Feature Values per Species (p10_linechart_means.png)

"This line chart plots the mean value of all 4 features for each species using 3 colored lines. Virginica's line is highest overall. Setosa's line dips dramatically for petal features. This clearly shows that petal features are the best for species discrimination."

Plot 4 — Scatter Plot: Petal Length vs Petal Width (p10_scatter_petal.png)

"This is the most important visualization. It plots petal_length (X) vs petal_width (Y) colored by species. **Setosa (blue) is completely separated** from the other two — forming a tight cluster at the bottom-left. Versicolor and Virginica overlap slightly but are largely separable. This confirms petal dimensions are the most useful features for classification."

Outlier Summary (IQR Method):

"Using IQR method: sepal_length (0 outliers), **sepal_width (4 outliers)**, petal_length (0), petal_width (0). Sepal width has a few extreme values — visible in the box plot as dots outside the whiskers."

Q: What is the difference between a Histogram and a Box Plot?

A: A histogram shows the full frequency distribution of a variable across bins — you can see shape (skewed, normal, bimodal). A box plot shows a 5-number summary (min, Q1, median, Q3, max) and is better for comparing distributions across groups and spotting outliers.

Q: What does a bimodal histogram tell you?

A: A bimodal histogram has two peaks, indicating two distinct groups or populations in the data. In Iris, petal features are bimodal because setosa petals are much smaller than versicolor/virginica.

Q: Why is the scatter plot of petal_length vs petal_width the most useful?

A: Because setosa is completely linearly separable from the other two species using just these two features. This means even a simple classifier can achieve very high accuracy using only petal dimensions.

Q: What is `.melt()` in the Line Chart section?

A: `df.melt()` converts a wide-format DataFrame (one column per feature) to a long-format (one row per feature-species combination). This makes it easier to plot multiple lines on the same chart using seaborn's `hue` parameter.

Q: What does `load_iris(as_frame=True)` do?

A: It loads the Iris dataset and returns it as a pandas DataFrame directly (instead of a numpy array), making it easier to work with column names.

✓ PRACTICAL 11 (A11.java) — Hadoop MapReduce WordCount

What This Practical Does

Implements the classic **MapReduce WordCount** program in Java using Hadoop — counts how many times each word appears in an input text file.

How to Explain (No Running Required — Theory + Steps)

Input used:

Hello World Hello Hadoop is great Hadoop

Expected Output:

Hadoop 2

```
Hello  2
World  1
great  1
is     1
```

Architecture Explanation

"Sir, the MapReduce program has 3 main components:"

1. Mapper Class

"The Mapper reads input line by line. For each line, it uses `StringTokenizer` to split it into individual words. For every word, it **emits a key-value pair (word, 1)**. So for 'Hello World Hello', it emits: `(Hello, 1)`, `(World, 1)`, `(Hello, 1)`."

2. Shuffle & Sort (automatic — done by Hadoop)

"Hadoop automatically groups all emitted pairs by key. So all `(Hello, 1)` pairs are grouped together: `(Hello, [1, 1])`. This is called the Shuffle and Sort phase."

3. Reducer Class

"The Reducer receives `(word, [list of 1s])` for each unique word. It sums all the 1s to get the total count and **emits (word, total)**. So `(Hello, [1, 1])` becomes `(Hello, 2)`."

4. Main Method (Driver)

"The main method configures the job — sets Mapper class, Reducer class, output key type (`Text`), output value type (`IntWritable`), input path, and output path. Then submits the job to Hadoop."

Terminal Commands to Explain

Step 1 - Create input

```
mkdir -p ~/wordcount/input
```

```
echo "Hello World Hello Hadoop is great Hadoop" > ~/wordcount/input/data.txt
```

Step 2 - Compile

```
hadoop com.sun.tools.javac.Main WordCount.java
```

```
jar cf WordCount.jar WordCount*.class
```

Step 3 - Run MapReduce job

```
hadoop jar WordCount.jar WordCount ~/wordcount/input ~/wordcount/output
```

Step 4 - View output

```
hadoop fs -cat ~/wordcount/output/part-r-00000
```

"The output file is named `part-r-00000` — 'r' stands for Reducer. The words are sorted alphabetically because Hadoop sorts keys before sending to the Reducer."

? Counter Questions & Answers

Q: What is Hadoop? What is MapReduce?

A: Hadoop is an open-source framework for distributed storage and processing of large datasets. MapReduce is the programming model it uses — Map phase transforms input data into key-value pairs, Reduce phase aggregates them.

Q: What is HDFS?

A: HDFS (Hadoop Distributed File System) is Hadoop's storage system. It splits large files into blocks (default 128MB) and distributes them across multiple nodes with replication for fault tolerance.

Q: What is LongWritable and IntWritable in Hadoop?

A: Hadoop uses its own serializable data types. `LongWritable` is Hadoop's equivalent of Java's `long`. `IntWritable` is Hadoop's equivalent of `int`. They are needed for efficient network serialization in distributed systems.

Q: Why does the output go to `part-r-00000`?

A: This naming convention comes from Hadoop — 'part' = partition, 'r' = reducer output, '00000' = partition number (0-indexed). If there were multiple reducers, you'd get `part-r-00001`, `part-r-00002` etc.

Q: Why do we need to delete the output folder before re-running?

A: Hadoop will refuse to write to an existing output directory to prevent accidental overwriting of results. You must delete it with `hadoop fs -rm -r ~/wordcount/output` before re-running.

Q: What is the difference between Map and Reduce?

A: Map transforms/filters data and emits intermediate key-value pairs. Reduce aggregates/summarizes the grouped pairs into final output. Map runs in parallel on all input splits; Reduce runs after all Maps complete.

Q: What is StringTokenizer?

A: A Java class that splits a string into tokens (words) based on delimiters (default: space, tab, newline). In WordCount, it splits each line into individual words.

Quick Summary Table — Practicals 6–11

Practica I	Topic	Key Output / Result
P6	Naive Bayes — Iris	100% accuracy , perfect confusion matrix (diagonal only)
P7	NLP + TF-IDF	6 preprocessing steps, TF-IDF='learning' in Doc2 = 0.0990
P8	Visualization — Titanic	3 plots: Histogram, Count Plot, Scatter; missing values handled
P9	Advanced Viz — Titanic	3 plots: Box Plot, Line Chart, Scatter; 500-row realistic dataset
P10	Visualization — Iris	4 plots; sepal_width has 4 outliers; setosa perfectly separable in scatter
P11	Hadoop MapReduce	WordCount Java — Hello=2, Hadoop=2, World=1, great=1, is=1

Exam tip: For P8, P9, P10 — always describe **what the axes represent, what the colors mean**, and most importantly **what conclusion you draw** from each plot. Examiners love it when you say *"this plot shows that..."* rather than just describing how to make it.